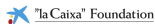


From Community Gardens to Rooftops: Practices, Innovation, and Regulations to Enhance Barcelona's Food System

Diego Maximiliano Macall

06-11-2025



Presentation Outline

- 1 Introduction
- 2 Hypothesis and Research Questions
- 3 Agronomic and Organizational Aspects of Barcelona's Community Gardens
- 4 A food system innovation: Vegetable production in rooftop greenhouses in Barcelona
- 5 Analysis of Regulations Governing the Scale Up of High Technology Urban Rooftop Horticulture in Barcelona
- 6 Thesis Discussion & Conclusion
- 7 Acknowledgements

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- Municipal governments around the world are not passive. They have increasingly positioned themselves as active players in food system transformation.

Barcelona as a Laboratory for Urban Agriculture: Context, Constraints, and Governance.

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Barcelona in Context Context

Table 1: Comparative Characteristics of Cities at a Similar Latitude

City	Barcelona	Chicago	Rome	Tashkent
Latitude	41°38'N	41°82'N	41°90'N	41°31'N
2025 Population (Millions)	1.6	2.7	2.75	3
Area (km ²)	101	600	1,285	438
Population Density (Inhabitants/km ²)	16,144	4,166	2,220	6,942
Green Space per Capita (m ² /Inhabitant)	7	14.2	39	10
Annual Sunshine Hours	2,683	2,508	2,473	2,820
Annual Mean Total Precipitation (mm)	582	937	804	419
Participation in the MUFPP	Yes	Yes	Yes	No
Dedicated Urban Agriculture Policies	Yes	Yes	Yes	Yes

Greenery Density in Barcelona Districts

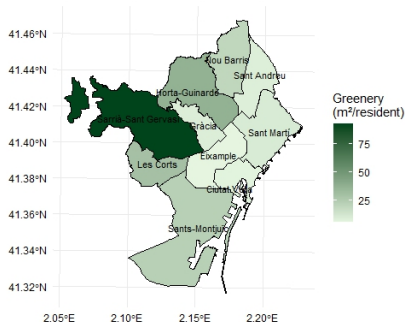


Figure 1: Greenery density (m^2 per resident) across Barcelona districts.
Source: Author based on Gerència d'Àrea d'Ecologia Urbana et al. (2021).

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- 1 What vegetables are currently cultivated in Barcelona's community gardens (CGs), and what production methods are employed in them and by whom?
- 2 What proportion of Barcelona's most demanded vegetables could technically be produced on the city's available rooftop area using high technology UH?
- 3 What do Barcelona's existing legal and regulatory frameworks entail for rooftop horticulture, and who ultimately holds the key to the expansion of high technology UH across the city?

Agronomic and Organizational Aspects of Barcelona's Community Gardens

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- This study examines the demographics, organizational structures, and agronomic practices of a sample of CGs within Barcelona's municipal boundaries.

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- Questionnaire (30 questions), comprising 11 multiple-choice and 19 open-ended questions.
- Specialized assessment tool used to visually document five key physical characteristics of the CGs visited during the study.

Results

- A total of 18 gardens, with varying stated main purposes or objectives were visited between July and November 2022.

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 - Agronomic Aspects of CGs Visited

Results

Table 2: Crops grown in Barcelona's community gardens

Common name	Latin name	Conventional yield (kg/m ²)	Organic yield (kg/m ²)
Basil	<i>Ocimum basilicum</i>	360 leaves per plant	219 leaves per plant
Bell pepper	<i>Capsicum annuum</i>	22.5	15.3
Cucumber	<i>Cucumis sativus</i>	27.5	18.7
Eggplant	<i>Solanum melongena</i>	23.7	16.6
Lettuce	<i>Lactuca sativa</i>	22.6	15.4
Onion	<i>Allium cepa</i>	31.3	21.3
Pumpkin	<i>Cucurbita maxima</i>	35.6	24.2
Squash	<i>Cucurbita</i>	29.1	19.8
Tomato	<i>Solanum lycopersicum</i>	33.0	22.4
Zucchini	<i>Cucurbita pepo</i>	29.5	20.0

Results



BNUG 2 (A)

BNUG 2 (B)

Figure 2: Polyculture in Barcelona's CGs.

Results



CIV 1

BNUG 8

BNUG 4

Figure 3: Elevated beds or Tables for Horticultural Production.

Discussion & Conclusion

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- “Amateur” labor is insufficient to sustain a productive urban food system (Mcdougall et al., 2020).

A food system innovation: Vegetable
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Introduction

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- Scaled up to a city-wide level, the employment of RTGs could enhance a city's vegetable self-sufficiency because they provide the environmental means through which these can be grown directly within cities.
- This study empirically measures the production potential of three fresh consumed vegetables in Barcelona within RTGs.

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- Together, these three vegetables represented 31% of total Barcelonan per capita consumption of fresh (not cooked or processed) vegetables in 2022.

Materials and Methods

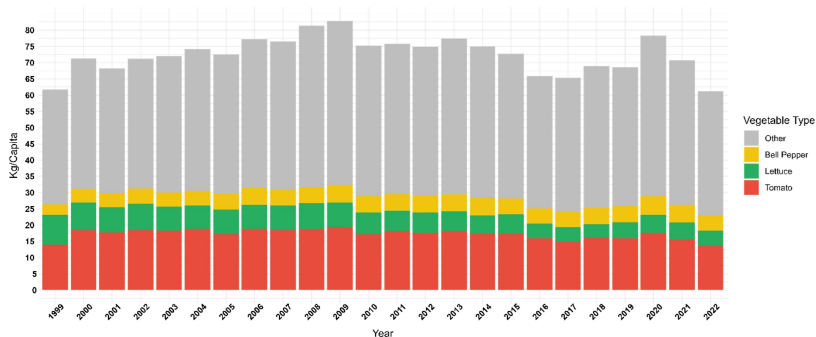


Figure 4: Catalanian annual per capita tomato, bell pepper, and lettuce consumption 1999-2022.

Source: Author based on MAPA (2024).

Materials and Methods

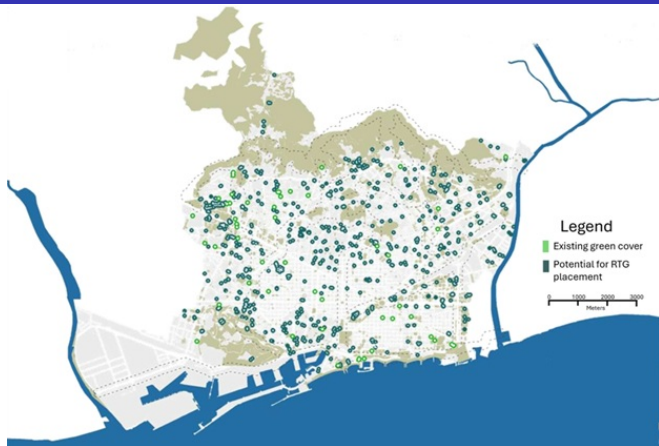


Figure 5: Spatial distribution of potential RTG sites and existing green cover in Barcelona.

Source: Adapted from AEUB (2010, p. 180).

RTG Tomato, Bell Pepper, and Lettuce Yield

Table 3: Maximum and minimum crop yields within RTGs.

Crop	Season	Max yield (g/plant)	Min yield (g/plant)	MC simulated yield per plant (g/plant)	Yield (kg/m ²)
Tomato	Wi-Sp-Su	8,463	5,179	6,821	13.6
Bell Pepper	Au-Wi	3,150	750	1,949	3.9
Lettuce	Sp	379	174	277	0.6

Source: Rufí-Salís et al. (2020)

RTG Establishment and Crop Allocation

$$\begin{aligned} \max R &= 9.82 x_1 + 3.47 x_2 + 0.28 x_3 \\ \text{s.t. } x_1 + x_2 + x_3 &\leq 650,000 & (1) \\ x_1 &= 390,000 & (2) \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

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- x_1 , x_2 , and x_3 represent square meters allocated to the cultivation of tomatoes, bell peppers, and lettuce respectively.

Water Requirements

- Rufí-Salís et al. (2020) show how much water is needed to produce the vegetables considered in this study.

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 - 78 t of lettuce requires 4,212 m³ of water to produce

Results

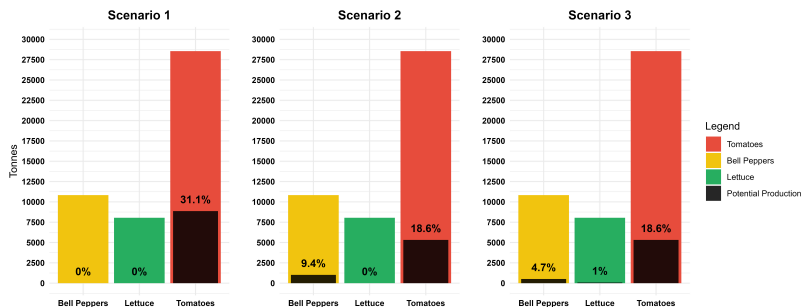


Figure 6: Average total tomato, bell pepper, and lettuce consumption between 1999-2022 in contrast to total production of these crops within 65 ha of RTGs

Discussion & Conclusion

- Through the scaling-up of UH within RTGs to a city-wide level, it is possible to supply a considerable amount of the tomatoes, bell peppers, and lettuce consumed by Barcelonans.

Production Sensitivity Analysis

Table 4: Total tomato, bell pepper, and lettuce production attainable through 90 m² RTGs.

Crop	Yield (kg/m ²)	Total buildings	Production per single RTG unit	Total production (t)
Tomato	13.6	437	1,224	535
Bell Pepper	3.9	437	351	153
Lettuce	0.6	437	54	24

Discussion & Conclusion

- This crop production system can contribute significantly to meeting local vegetable demand.
- The substantial production can be achieved in a mere 65 ha of rooftop area.
- This study establishes a potential production baseline.

Analysis of Regulations Governing the Scale Up of High Technology Urban Rooftop Horticulture in Barcelona

- This study identifies the laws and ordinances regulating rooftop use in Barcelona and analyzes the institutional statements within these them that either directly or indirectly facilitate or restrict the installation of high technology UH.

Barcelona's Special Status

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Institutional Grammar characteristics used in this study

Table 5: Institutional Grammar characteristics used in this study

Component	Acronym	Specification
Attribute	A	The individual(s), group(s), or organization(s) that carry out the alm.
oBject	B	Animate or inanimate part of the statement that is the receiver of an action.
Deontic	D	The prescriptive component of an institutional statement that specifies what is allowed, required, or prohibited.
alm	I	The alm specifies the objective or action described by the institutional statement to which the Deontic refers.
Condition	C	Part of the statement that modifies the alm (in temporal or spatial terms) or describes how the action recognized in the alm is to occur.
Or else	O	Punitive action if the rule is not adhered to.

Source: Based on Crawford and Ostrom (1995), Ostrom (2005), and Siddiki et al. (2011).

Materials and Methods

- Three steps:

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- 1 Identification of Relevant Regulations.

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- 1 Identification of Relevant Regulations.
- 2 Extraction and Translation of Relevant Articles. To ensure consistency, identified Deontic terms in these Articles (e.g., permitted, accepted, allowed, must, prohibited, authorized, or not obligated) were standardized during the translation process to align with common usage in the literature e.g., “must,” “may,” or “must not,” following the framework reestablished by Frantz and Siddiki (2022).

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- 3 Analyzing Institutional Statements Using the IGT Framework.

Key institutions governing Barcelona's rooftop area

Table 6: Key institutions governing Barcelona's rooftop area.

Institution	Aim	Article(s)
Spanish Civil Code (Law/Royal Decree)	Defines Spain's civil law framework, covering property rights, contracts, family relations, and inheritance.	396
Horizontal Property Law (Law 49/1960)	Governs ownership and management of buildings with multiple owners and shared common areas.	17.4
Codi Civil de Catalunya (Civil Code of Catalonia)	Regulates property, inheritance, family law, contracts, and co-ownership within Catalonia.	553-26(c), 553-43
Ordinance on the Uses of the Urban Landscape of the City of Barcelona	Safeguards the city's harmonious landscape, regulating private uses that could distort its historic, artistic, or traditional ensembles.	74, 76

Source: Author based on Ajuntament de Barcelona (2024).

Table 7: Summary statistics of policy instruments deconstructed

Legal instrument	Num of articles dissected	Num of institutional statements extracted
Horizontal Property (Law 49/1960)	1	19
Civil Code of Catalonia	2	14
Ordinance on the uses of the urban landscape of the city of Barcelona (Norm)	2	64

Results

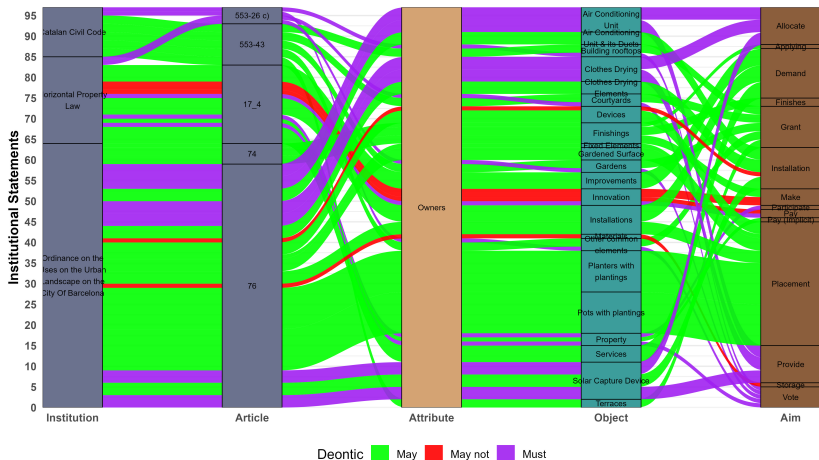


Figure 7: Dissection of Policy Instruments and relevant Articles to the integration of UH onto Barcelona's buildings

Results

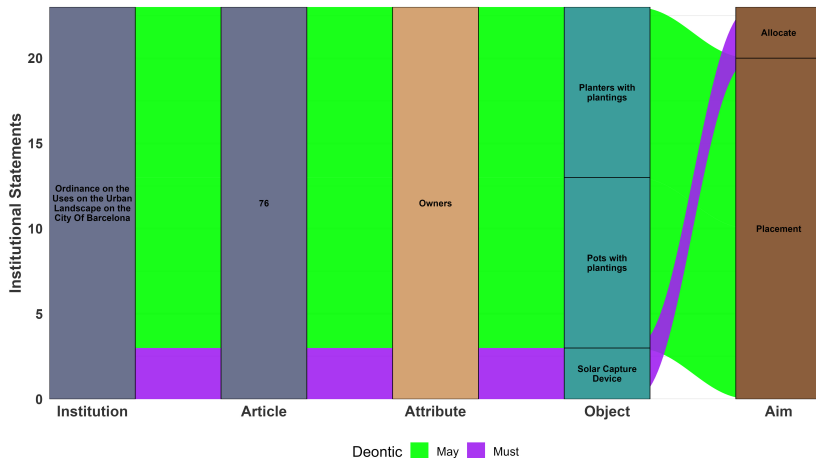


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Discussion & Conclusion

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Discussion & Conclusion

- Barcelona's municipal authorities play a crucial role in legitimizing and formalizing UH as a recognized and permissible use of rooftop spaces.
- However, its scalability ultimately depends on building owners, as they hold the power to enact modifications to the rooftops of their properties.

Thesis Discussion & Conclusion

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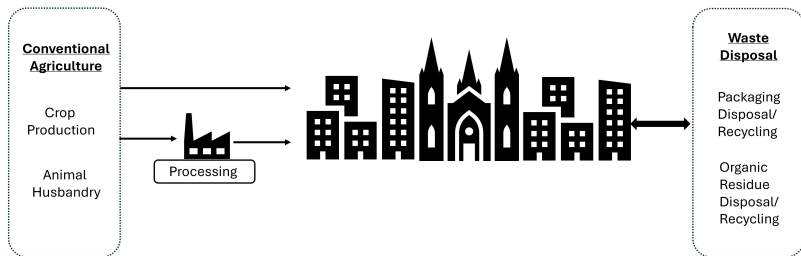


Figure 8: Integration of high technology UH into Barcelona's urban fabric.

Thesis Discussion

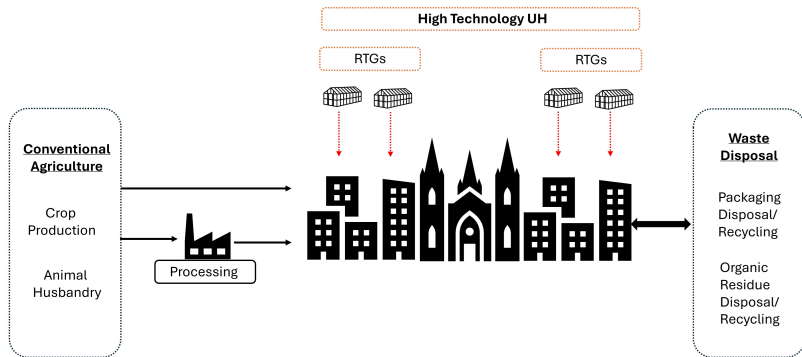


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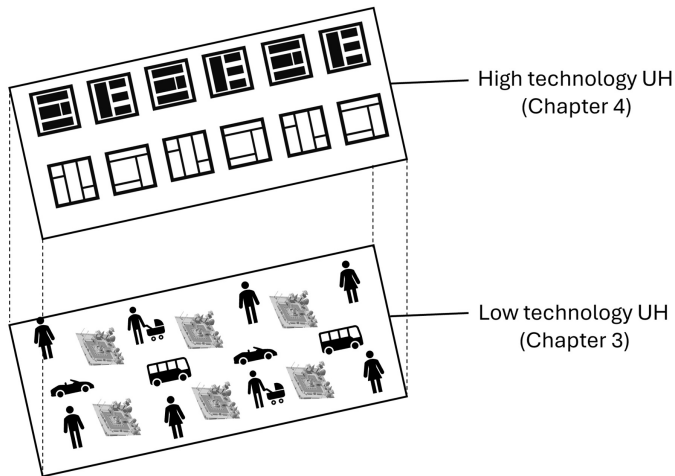


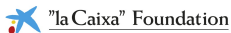
Figure 9: Dimensional Analysis of UH. The top panel shows the city's rooftop area, and the bottom one shows the ground level area where UH could be located.

Thesis Conclusion

- Barcelona's CGs are best left to provide social benefits.
- Barcelona's rooftops can be used to fundamentally transform the city's food system.
- Barcelonans hold the key to transforming their food city's food system.

Acknowledgements

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